

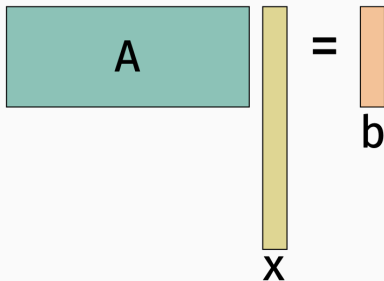
CS-GY 9223 I: Lecture 13

Compressed Sensing + Sparse Recovery

NYU Tandon School of Engineering, Prof. Christopher Musco

BASIC PROBLEM SETUP

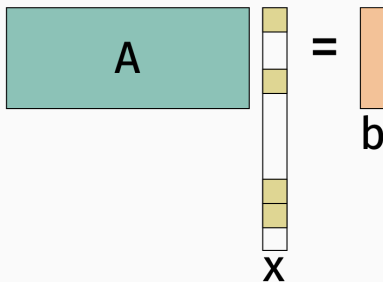
Underdetermined linear regression: Given $A \in \mathbb{R}^{m \times n}$ with $m < n$, $b \in \mathbb{R}^m$. Solve $Ax = b$ for x .


$$A x = b$$

- Infinite possible solutions x . In general, impossible to recover parameter vector.

Underdetermined linear regression: Given $A \in \mathbb{R}^{m \times n}$ with $m < n$, $b \in \mathbb{R}^m$. Solve $Ax = b$ for x .

- Assume x is k -sparse for small k . $\|x\|_0 = k$.



- In many cases can recover x with $\ll n$ rows. In fact, often $\sim O(k)$ suffice.
- Need additional (strong) assumptions about A !

QUICK ASIDE

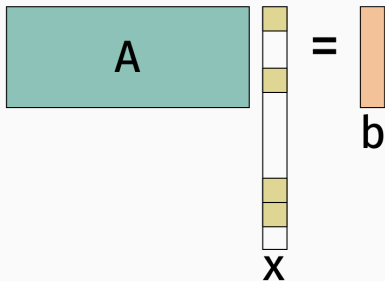
- In the past, we have thought about $\mathbf{A}$'s rows as data drawn from some universe/distribution:

	bedrooms	bathrooms	sq.ft.	floors	list price	sale price
home 1	2	2	1800	2	200,000	195,000
home 2	4	2.5	2700	1	300,000	310,000
.	.	.	.	.	.	.
.	.	.	.	.	.	.
.	.	.	.	.	.	.
home n	5	3.5	3600	3	450,000	450,000

- In many settings, we will get to choose $\mathbf{A}$'s rows. I.e. each $b_i = \mathbf{x}^T \mathbf{a}_i$ for some vector $\mathbf{a}_i$ that we select.
- In this setting, we often call b_i a linear measurement of $\mathbf{x}$ and we call $\mathbf{A}$ a measurement matrix.

ASSUMPTIONS ON MEASUREMENT MATRIX

When should this problem be difficult?



The diagram illustrates the linear equation $Ax = b$. On the left, a teal rectangle labeled A represents the measurement matrix. To its right is a vertical column labeled x , representing the unknown vector. This column is divided into eight segments: the first, third, fourth, sixth, and seventh segments are yellow, while the second, fifth, and eighth segments are white. To the right of the column x is an equals sign $=$, followed by a vertical orange rectangle labeled b , representing the observed vector.

Many ways to formalize our intuition

- **A** has Kruskal rank r . All sets of r columns in **A** are linearly independent.
 - Recover vectors **x** with sparsity $k = r/2$.
- **A** is μ -incoherent. $|\mathbf{A}_i^T \mathbf{A}_j| \leq \mu \|\mathbf{A}_i\|_2 \|\mathbf{A}_j\|_2$ for all columns $\mathbf{A}_i, \mathbf{A}_j$.
 - Recover vectors **x** with sparsity $k = 1/\mu$.
- **Focus today:** **A** obeys the Restricted Isometry Property.

Definition $((q, \epsilon)$ -Restricted Isometry Property)

A matrix $\mathbf{A}$ satisfies (q, ϵ) -RIP if, for all $\mathbf{x}$ with $\|\mathbf{x}\|_0 \leq q$,

$$(1 - \epsilon)\|\mathbf{x}\|_2^2 \leq \|\mathbf{Ax}\|_2^2 \leq (1 + \epsilon)\|\mathbf{x}\|_2^2.$$

- Johnson-Lindenstrauss type condition.
- $\mathbf{A}$ preserves the norm of all q sparse vectors, instead of the norms of a fixed discrete set of vectors, or all vectors in a subspace (as in subspace embeddings).

Theorem (ℓ_0 -minimization)

Suppose we are given $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} = \mathbf{A}\mathbf{x}$ for an unknown k -sparse $\mathbf{x} \in \mathbb{R}^n$. If $\mathbf{A}$ is $(2k, \epsilon)$ -RIP for any $\epsilon < 1$ then $\mathbf{x}$ is the unique minimizer of:

$$\min \|\mathbf{z}\|_0 \quad \text{subject to} \quad \mathbf{A}\mathbf{z} = \mathbf{b}.$$

- Establishes that information theoretically we can recover $\mathbf{x}$. Solving the ℓ_0 -minimization problem is computationally difficult, requiring $O(n^k)$ time. We will address faster recovery later in the lecture.

Proof:

Important note: Robust versions of this theorem and the others we will discuss exist. These are much more important practically. Here's a flavor of a robust result:

- Suppose $\mathbf{b} = \mathbf{A}(\mathbf{x} + \mathbf{e})$ where $\mathbf{x}$ is k -sparse and $\mathbf{e}$ is dense but has bounded norm.
- Recover some k -sparse $\tilde{\mathbf{x}}$ such that:

$$\|\tilde{\mathbf{x}} - \mathbf{x}\|_2 \leq \|\mathbf{e}\|_1$$

or even

$$\|\tilde{\mathbf{x}} - \mathbf{x}\|_2 \leq O\left(\frac{1}{\sqrt{k}}\right) \|\mathbf{e}\|_1.$$

We will not discuss robustness in detail, but it is a big part of what has made compressed sensing such an active research area in the last 20 years. Non-robust compressed sensing results have been known for a long time:

Gaspard Riche de Prony, *Essay experimental et analytique: sur les lois de la dilatabilite de fluides elastique et sur celles de la force expansive de la vapeur de l'alcool, a differentes temperatures*. Journal de l'Ecole Polytechnique, 24–76. **1795**.

What matrices satisfy this property?

- Random Johnson-Lindenstrauss matrices (Gaussian, sign, etc.) with $m = O(\frac{k \log(n/k)}{\epsilon^2})$ rows are $(O(k), \epsilon)$ -RIP.

Some real world data may look random, but this is also a useful observation algorithmically when we want to design A.

APPLICATION: HEAVY HITTERS IN DATA STREAMS

Suppose you view a stream of numbers in $1, \dots, n$:

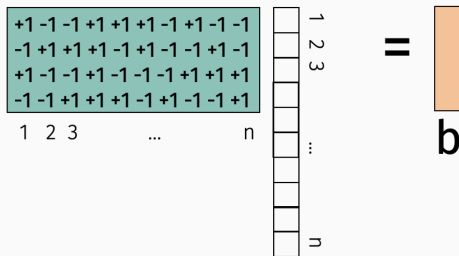
4, 18, 4, 1, 2, 24, 6, 4, 3, 18, 18, $\dots$

After some time, you want to report which k items appeared most frequently in the stream.

E.g. Amazon is monitoring web-logs to see which product pages people view. They want to figure out which products are viewed most frequently. $n \approx 500$ million.

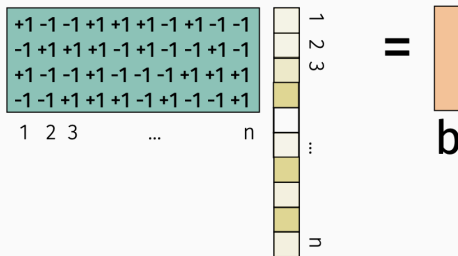
How can you do this quickly in small space?

APPLICATION: HEAVY HITTERS IN DATA STREAMS



- Every time we receive a number i in the stream, add column A_i to b .

APPLICATION: HEAVY HITTERS IN DATA STREAMS



- At the end $\mathbf{b} = \mathbf{A}\mathbf{x}$ for an approximately sparse $\mathbf{x}$ if there were only a few “heavy hitters”. Recover $\mathbf{x}$ from $\mathbf{b}$ using a sparse recovery method (like ℓ_0 minimization).

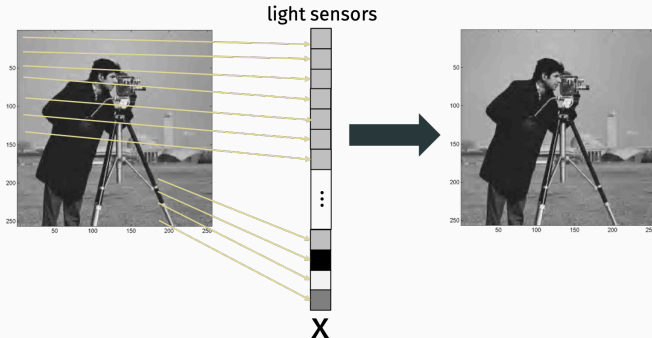
How about when there are insertions or deletions?

insert(4), insert(18), remove(4), insert(1), insert(2), remove(2) ...

E.g. Amazon is monitoring what products people add to their “wishlist” and wants a list of most tagged products. Wishlists can be changed over time, including by removing items.

APPLICATION: SINGLE PIXEL CAMERA

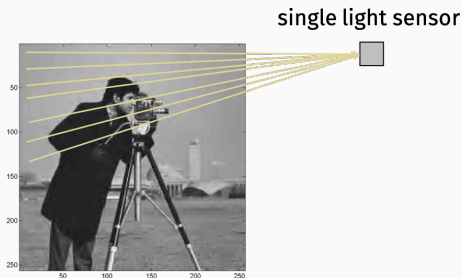
Typical acquisition of image by camera:



Requires one image sensor per pixel captured.

APPLICATION: SINGLE PIXEL CAMERA

Compressed acquisition of image:

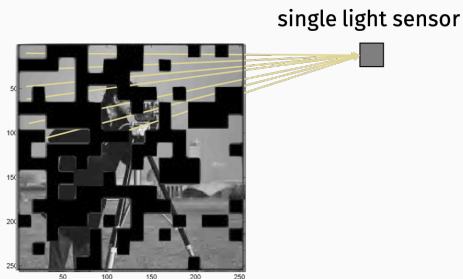


$$p = \sum_{i=1} x_i = \begin{bmatrix} 1 & 1 & \dots & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Does not provide very much information about the image.

APPLICATION: SINGLE PIXEL CAMERA

But several random linear measurements do!



$$p = \sum_{i=1} R_i x_i = \begin{bmatrix} 0 & 1 & 0 & 0 & \dots & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Applications in:

- Imaging outside of the visible spectrum (more expensive sensors).
- Microscopy.
- Other scientific imaging.

Compressed sensing theory does not exactly describe the problem, but has been very valuable in modeling it.

RESTRICTED ISOMETRY PROPERTY

Definition $((q, \epsilon)$ -Restricted Isometry Property)

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Uniformly subsampled Fourier matrices with $m \sim O\left(\frac{k \log^2 k \log n}{\epsilon^2}\right)$ rows $(O(k), \epsilon)$ -RIP. [Haviv, Regev, 2016].

Improves on a long line of work: Candès, Tao, Rudelson, Vershynin, Cheraghchi, Guruswami, Velingker, Bourgain.

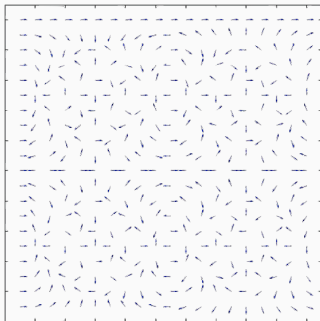
You have seen some of the tools used prove this when we proved that a subsampled Hadamard matrix, which is a type of Fourier matrix, can be used to give a JL guarantee.

THE DISCRETE FOURIER MATRIX

The $n \times n$ discrete Fourier matrix $\mathbf{F}$ is defined:

$$F_{j,k} = e^{\frac{-2\pi i}{n}j \cdot k}$$

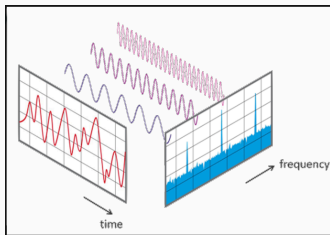
Recall that $e^{\frac{-2\pi i}{n}j \cdot k} = \cos(2\pi jk/n) - i \sin(2\pi jk/n)$.



Set $\mathbf{A}$ to contain a random $\tilde{O}(k \log n)$ rows of this matrix.

THE DISCRETE FOURIER MATRIX

$\mathbf{F}\mathbf{x}$ is the Discrete Fourier Transform of the vector $\mathbf{x}$ (what an FFT computes).



Decomposes $\mathbf{x}$ into different frequencies: $[\mathbf{F}\mathbf{x}]_j$ is the component with frequency j/n .

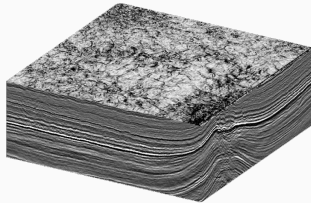
Because $\mathbf{F}^*\mathbf{F} = \mathbf{I}$, $\mathbf{F}^*\mathbf{F}\mathbf{x} = \mathbf{x}$, so we can recover $\mathbf{x}$ if we have access to its DFT. $\mathbf{F}\mathbf{x}$.

If $\mathbf{A}$ is a subset of q rows from $\mathbf{F}$, then $\mathbf{Ax}$ is a subset of random frequency components from $\mathbf{x}$'s discrete Fourier transform.

In many scientific applications, we can collect entries of $\mathbf{Fx}$ one at a time for some unobserved data vector $\mathbf{x}$.

Warning: very cartoonish explanation of very complex problem.

Understanding what material is beneath the crust:



Think of vector $\mathbf{x}$ as scalar values of the density/reflectivity in a single vertical core of the earth.

How do we measure entries of Fourier transform $\mathbf{F}\mathbf{x}$?

APPLICATION: GEOPHYSICS

Vibrate the earth at different frequencies! And measure the response.



Vibroseis Truck

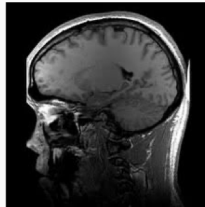
Can also use airguns, controlled explosions, vibrations from drilling, etc. The fewer measurements we need from F_x , the cheaper and faster our data acquisition process becomes.

Killer app: Oil Exploration.

APPLICATION: GEOPHYSICS

Warning: very cartoonish explanation of very complex problem.

Medical Imaging (MRI, Ultrasound, etc.)



Vector $\mathbf{x}$ here is a 2D image. Everything works with 2D Fourier transforms.

How do we measure entries of Fourier transform $\mathbf{F}\mathbf{x}$?

APPLICATION: GEOPHYSICS

Blast the body with sound waves or ultrasonic waves of varying frequencies.



The fewer measurements we need from **F_x**, the faster we can acquire and image.

- Especially important when trying to capture something moving (e.g. lungs, baby, child who can't sit still).
- Can also cut down on power requirements (which for MRI

RESTRICTED ISOMETRY PROPERTY

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Lots of other random matrices satisfy RIP as well.

One major theoretical question is if we can deterministically construct good RIP matrices. Interestingly, if we want $(O(k), O(1))$ RIP, we can only do so with $O(k^2)$ rows (now very slightly better – thanks Bourgain et al.).

Whether or not a linear dependence on k is possible with a deterministic construction is unknown.

Theorem (ℓ_0 -minimization)

Suppose we are given $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{b} = \mathbf{A}\mathbf{x}$ for an unknown k -sparse $\mathbf{x}$. If $\mathbf{A}$ is $(2k, \epsilon)$ -RIP for any $\epsilon < 1$ then $\mathbf{x}$ is the unique minimizer of:

$$\min \|\mathbf{z}\|_0 \quad \text{subject to} \quad \mathbf{A}\mathbf{z} = \mathbf{b}.$$

Algorithm question: Can we recover $\mathbf{x}$ using a faster method?
Ideally in polynomial time.

Convex relaxation of the ℓ_0 minimization problem:

Problem (Basis Pursuit, i.e. ℓ_1 minimization.)

$$\min_{\mathbf{z}} \|\mathbf{z}\|_1 \quad \text{subject to} \quad \mathbf{A}\mathbf{z} = \mathbf{b}.$$

- Objective is convex:
- Optimizing over convex set:

What is one method we know for solving this problem?

Equivalent formulation:

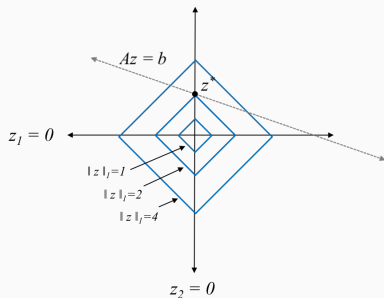
Problem (Basis Pursuit Linear Program.)

$$\min_{\mathbf{w}, \mathbf{z}} \mathbf{1}^T \mathbf{w} \quad \text{subject to} \quad \mathbf{Az} = \mathbf{b}, -\mathbf{w} \leq \mathbf{z} \leq \mathbf{w}.$$

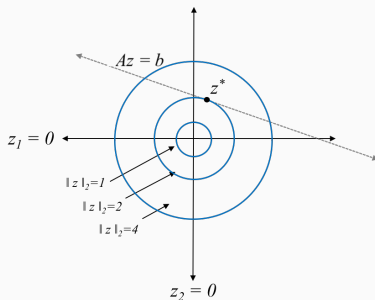
Can be solved using any algorithm for linear programming. An Interior Point Method will run in at worst $\sim O(n^{3.5})$ time.

BASIS PURSUIT INTUITION

Suppose $\mathbf{A}$ is 2×1 , so $\mathbf{b}$ is just a scalar and $\mathbf{z}$ is a 2-dimensional vector.



Vertices of level sets of ℓ_1 norm correspond to sparse solutions.



This is not the case e.g. for the ℓ_2 norm.

Theorem

If $\mathbf{A}$ is $(3k, \epsilon)$ -RIP for $\epsilon < .17$ and $\|\mathbf{x}\|_0 = k$, then $\mathbf{z}^ = \mathbf{x}$ is the unique optimal solution of the Basis Pursuit LP).*

Similar proof to ℓ_0 minimization:

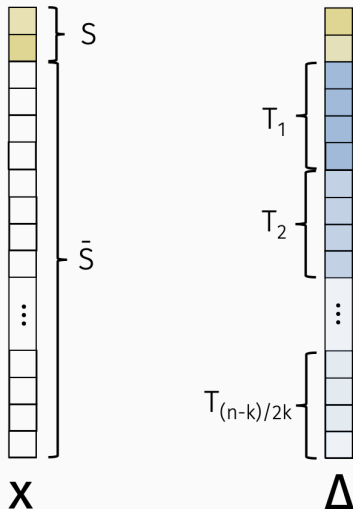
- By way of contradiction, assume $\mathbf{x}$ is not the optimal solution. Then there exists some non-zero Δ such that:
 - $\|\mathbf{x} + \Delta\|_1 \leq \|\mathbf{x}\|_1$
 - $\mathbf{A}(\mathbf{x} + \Delta) = \mathbf{A}\mathbf{x}$. i.e. $\mathbf{A}\Delta = 0$.

Difference is that we can no longer assume that Δ is sparse.

Only one tool needed:

For any q -sparse vector $\mathbf{w}$, $\|\mathbf{w}\|_2 \leq \|\mathbf{w}\|_1 \leq \sqrt{q}\|\mathbf{w}\|_2$

Some definitions:



Claim 1: $\|\Delta_S\|_1 \geq \|\Delta_{\bar{S}}\|_1$

Claim 2: $\|\Delta_S\|_2 \geq \sqrt{2} \sum_{j \geq 2} \|T_j\|_2$:

Finish up proof by contradiction:

A lot lot of interest in developing even faster algorithms that avoid using the “heavy hammer” of linear programming and run in even faster than $O(n^{3.5})$ time.

- **Iterative Hard Thresholding:** Looks a lot like projected gradient descent. Solve $\min_z \|\mathbf{A}z - \mathbf{b}\|$ while continually projecting z back to the set of k -sparse vectors. Runs in time $\sim O(nk \log n)$ for Gaussian measurement matrices and $O(n \log n)$ for subsampled Fourier matrices.
- Other “first order” type methods: Orthogonal Matching Pursuit, CoSaMP, Subspace Pursuit, etc.

When $\mathbf{A}$ is a subsampled Fourier matrix, there are now methods that run in $\underline{O(k \log^c n)}$ time [Hassanieh, Indyk, Kapralov, Katabi, Price, Shi, etc. 2012+].

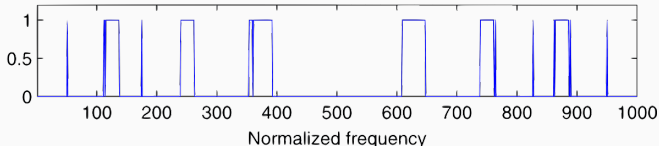
Hold up...

SPARSE FOURIER TRANSFORM

Corollary: When $\mathbf{x}$ is k -sparse, we can compute the inverse Fourier transform $\mathbf{F}^*\mathbf{F}\mathbf{x}$ of $\mathbf{F}\mathbf{x}$ in $O(k \log^c n)$ time!

- Randomly subsample $\mathbf{F}\mathbf{x}$.
- Feed that input into our sparse recovery algorithm to extract $\mathbf{x}$.

Fourier and inverse Fourier transforms in sublinear time when the output is sparse.



Applications in: Wireless communications, GPS, protein imaging, radio astronomy, etc. etc.